

Rohail Alam

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EXPERIENCE

AI Engineer

Oct 2025

BalanX Bio

Ames, IA

- **Led development of the emotion intelligence subsystem** for an emotionally aware LLM-powered avatar, architecting a real-time multimodal emotion recognition pipeline integrating speech, text, and facial cues.
- Trained a multimodal emotion classification model using MELD and CREMA-D, implementing **cross-attention fusion** with **LoRA-based fine-tuning** to obtain a classification accuracy of 74%, **beating the baseline** by 13%
- **AWS deployment** set up by building Docker images and orchestrating them via Amazon ECR and ECS. Inter-layer communication established through **Redis (Elasticache)** with analysis through **Redshift**.
- Optimized inference pipeline to achieve **sub-second latency**, enabling real-time interactions with the deployed app.

Data Science Intern

May 2023 – Nov 2023

Lincode Labs

Remote

- Optimized electronic board defect inspection cycle time for PCBA project **from 10 minutes to under 30 seconds** using Celery multiprocessing and load balancing with NGINX.
- Built and documented **RESTful Django APIs** using MongoDB collections for inference results, with **Postman** testing
- Trained a **YOLOv8** model (Average F1 of 86%) on an annotated dataset with synthesized defects to avoid the cost incurred in intentionally damaging a functional board.
- Containerized workflows with **Docker** to parallelize and automatically scale up/down based on available resources.

PROJECTS

Real-Time LLM Advisory System for Social Deduction

- A real-time LLM backed helper for the social deduction game '**Town of Salem**'.
- Built a custom **RAG pipeline** over data scraped from the official game wiki using **LangChain**, **ChromaDB** and embedding with a **sentence-transformers** model, utilized to obtain role information and common strategies.
- Wired an event bridge powered by the Tampermonkey browser extension to obtain live browser-side game events, prompting the LLM only at meaningful game events thereby ensuring **cost efficiency** for LLM calls.

Classifying Paper Relevancy based on Abstract

- **Fine-tuned DeBERTa** for document relevance classification using title + abstract inputs, achieving **88% F1-score** against the held-out test set organized in a **Kaggle Competition**
- Designed and executed end-to-end training workflows using **Azure ML**, including experiment tracking, GPU compute clusters, and scalable model training for NLP workloads.
- Utilized the Together.ai LLM API to **generate synthetic samples** for minority classes, applying **prompt engineering** techniques to preserve semantic and domain-specific information.

EDUCATION

Iowa State University

Aug. 2024 – December 2025

Masters of Science, Artificial Intelligence

Ames, IA

- **Coursework:** Computational Perception, ML, NLP, Industrial Research in AI, AI in Healthcare
- **Capstone:** ROS Implementation of Multi-objective Path Planning in Robots

SKILLS

AI/ML : Pytorch, LangChain, NLP, LLMs, Transformers, Finetuning, GPU Optimization, Multimodal AI

Programming : Python, C++, Java, Elm, SQL, MongoDB, ChromaDB, Competitive Coding (Leetcode, Codeforces)

Tools : AWS (CCP), MS Azure, Docker, REST APIs, Postman, Figma, Miro, Jira (Clickup)